

Topic List

Mathematical Statistics

2026 spring semester

As a general rule, students are expected to meet the following requirements.

- **Concepts** must be defined precisely and illustrated with examples. In some cases this part includes the description of a model; in such cases, students should be able to write down the model and state its assumptions.
- **Theorems** must be stated clearly and proved where required.
- **Procedures** must be known at the level of their general description and must be applicable in concrete situations.

1 Basic Statistical Concepts

Concepts

- Population
- Sample
- Data matrix
- Mean, empirical variance, corrected empirical variance
- Empirical distribution function

Theorems

- The Glivenko–Cantelli theorem.

Procedures

- Computing the empirical distribution function.

2 Estimation Theory

Concepts

- Point estimation
- Unbiasedness, asymptotic unbiasedness, bias, standard error, efficiency
- Consistency and strong consistency
- Interval estimation
- Confidence level

Theorems

- The sample mean is an unbiased estimator of the expected value; the standard error of the sample mean; the sample mean as a consistent estimator of the expected value.
- The empirical variance is an asymptotically unbiased estimator of the variance; the corrected empirical variance is an unbiased estimator of the variance.
- Properties of maximum likelihood estimators, both in finite-sample form where discussed and asymptotically.
- Confidence intervals for the expected value of a normal distribution when the standard deviation is known and when it is unknown.

Procedures

- Maximum likelihood estimation
- The method of moments

3 Introduction to Hypothesis Testing and Parametric Tests

Concepts

- Hypotheses: null hypothesis H_0 and alternative hypothesis H_1
- Significance level
- p -value
- Type I error
- Type II error
- Critical region and acceptance region
- Decision rules
- Test statistic
- One-sided and two-sided tests
- Independent-samples tests and paired-samples tests
- Large-sample tests

Theorems

For every parametric test, students should know the assumptions under which the test can be applied, the steps of the test, and the distribution of the test statistic under the null hypothesis. For Welch's test, the formula for the degrees of freedom is not required. In addition, students should know the following:

- derivation of the one-sample and two-sample z -test formulas;
- the formula for the one-sample t -test.

Procedures

- z -tests
- t -tests
- F -test
- Welch's test

4 Nonparametric Tests

Concepts

- Nonparametric test
- Ordinal variable and categorical variable

Theorems

- The Kolmogorov–Smirnov theorem.

Procedures

For the following tests, students should know the assumptions, the null hypotheses, the test statistics, and the asymptotic distributions of the test statistics.

- Kolmogorov–Smirnov tests
- χ^2 -tests
- Sign test
- Wilcoxon rank tests
- Mann–Whitney test

5 Multi-Sample and Exact Tests

Concepts

Students should understand the motivation for introducing the following concepts and procedures.

- Multi-sample tests
- Exact tests

Theorems

- No additional theorem is required in this topic beyond the statements and properties connected to the procedures below.

Procedures

For the following tests, students should know the assumptions, the null and alternative hypotheses, and the asymptotic distributions of the test statistics.

- ANOVA; here students should also know the derivation of the test statistic formula
- Welch ANOVA
- Repeated-measures ANOVA
- Kruskal–Wallis test
- Friedman test
- Fisher's exact test, including the Lady Tasting Tea problem
- Binomial test
- McNemar test

6 Linear Regression

Concepts

- Empirical covariance and correlation
- Theoretical regression
- The purpose of regression analysis
- Form of the linear regression model
- Dependent and independent variables
- Assumptions on the error terms

Theorems

- **Gauss–Markov theorem:** Under the assumptions of the linear regression model, the parameters estimated by the method of least squares are unbiased and have minimum variance among linear unbiased estimators.
- Definition and interpretation of R^2 .

Procedures

For the tests mentioned below, students should know the name of the test and the corresponding null hypothesis.

- Parameter estimation by the method of least squares: derivation in the case of one explanatory variable, and the formula in the case of several variables
- Regression coefficients, standardized coefficients, and prediction
- Goodness-of-fit diagnostics: derivation of the formula for R^2 ; for adjusted R^2 , the formula is not required, but students should know its purpose, motivation, and why it is introduced
- Diagnostics for multicollinearity: why multicollinearity is a problem, the definition of VIF, and at least one other diagnostic measure
- Model-fit diagnostics: significance tests for the model and for the coefficients, and diagnostic tests for the assumptions on the error terms

7 Time Series Analysis

Concepts and Models

- The concept of a time series and the main difficulties of time series modelling
- Dependence measures for theoretical time series: autocorrelation, partial autocorrelation in an informal sense, and correlogram
- Components of time series models:
 - trend
 - seasonality
 - cyclicity
 - error term
- Deterministic and stochastic models, with examples
- Deterministic model
- Exponential smoothing model without trend or seasonality
- MA(p) models and their properties
- AR(q) models and their properties

Theorems

- No separate theorem is required for this topic beyond the model properties discussed in class.

Procedures

For the following tests, students only need to know the null hypothesis, that is, what the test is used for.

- Filtering trends:
 - deterministic methods: regression and moving-average trend estimation;
 - stochastic trend filtering.
- Filtering seasonality:
 - removing deterministic seasonality.
- Diagnostics for error terms
- Ljung–Box tests
- Dickey–Fuller test

8 Statistical Pitfalls and Paradoxes

Students should be able to describe the following phenomena in general terms, recognize them in examples, and give examples of their own.

Misleading Practices

- Misleading data visualizations
- Reporting data without sufficient context
- Choosing an inappropriate descriptive statistic
- Applying statistical methods without checking assumptions, or omitting diagnostic checks
- Confusing correlation with causal interpretation
- Drawing conclusions from small samples
- p -hacking
- Sampling biases

Misinterpretations

- Prosecutor's fallacy
- The rare-disease problem and base-rate neglect
- Time-related bias
- Confusing absolute and relative frequencies
- Misuse of relative frequency when percentages are very small

Paradoxes and Phenomena

- Simpson's paradox
- Berkson's paradox
- Hand's paradox